

B747 Ramp BUD Magnet Cosmetic Standard-A3

Rev. A3
Magnet Vendor: Magsound
2025/08/04

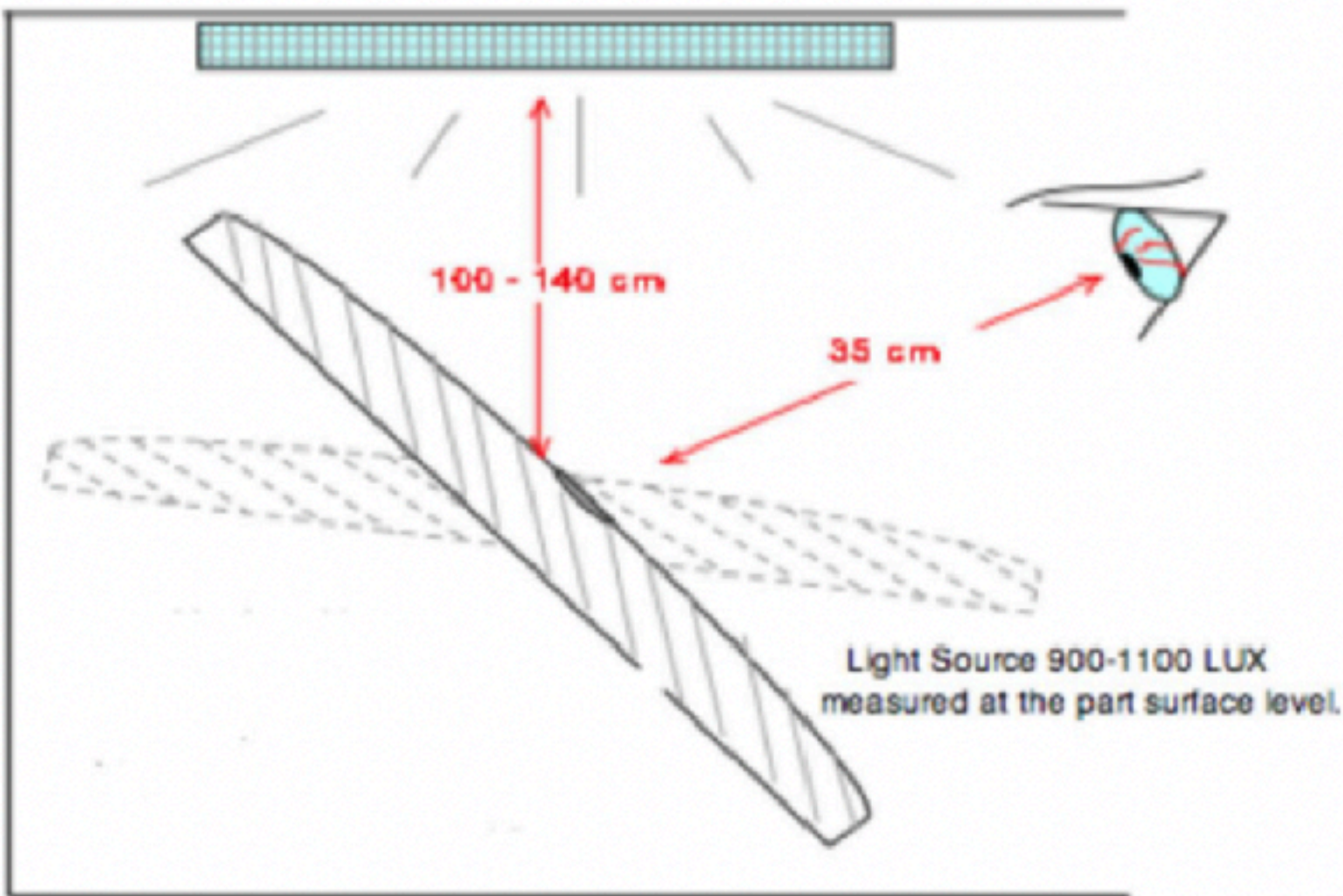
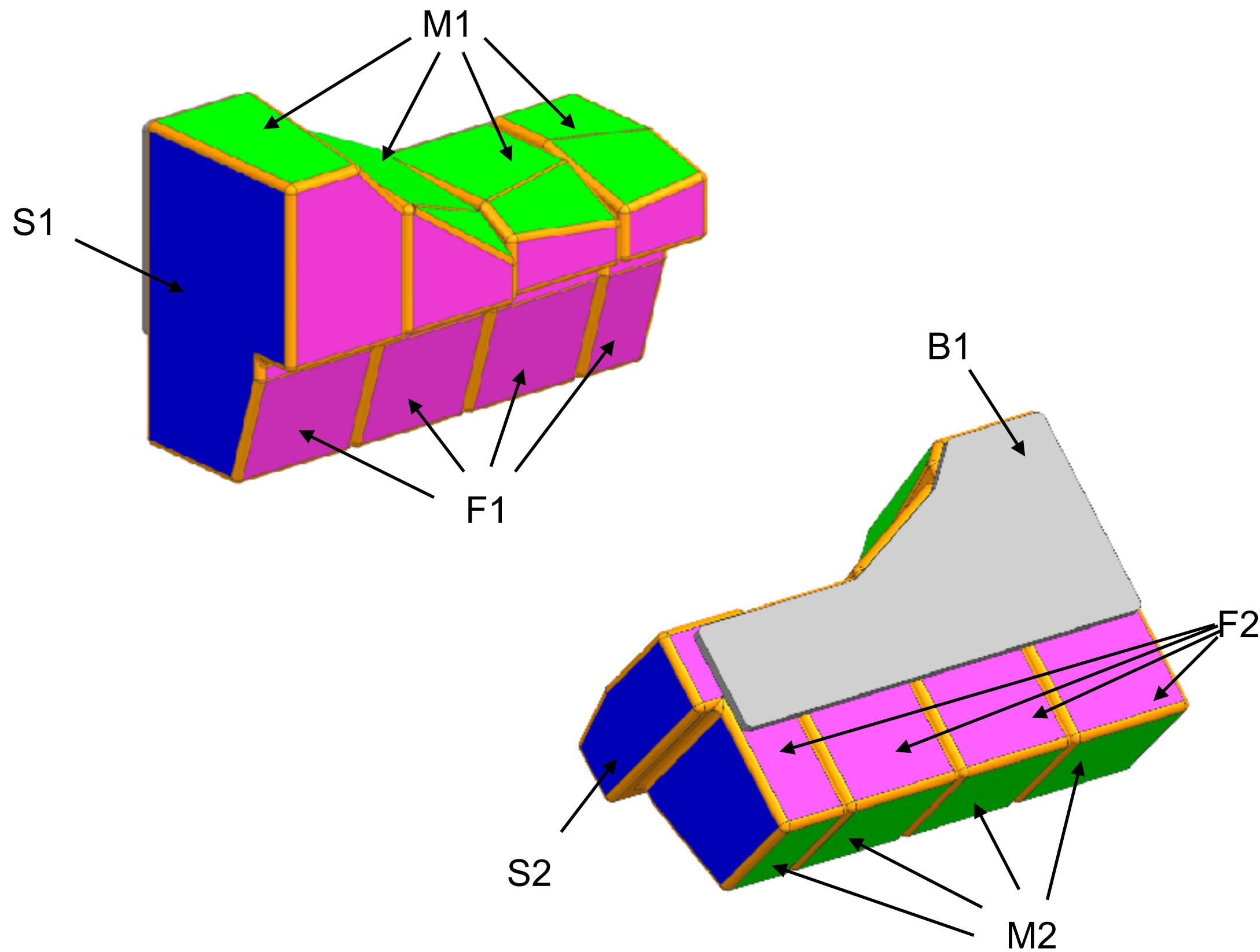


Fig. 1 Inspection criteria for opaque surfaces / all classes

- Lighting condition: 900-1100 Lux
- Magnification not to exceed 1X unless otherwise noted
- Surface to be evaluated at a distance of 35cm
- Part to be rotated freely for inspection
- Finger Feel base on check with glove on.
- Vertical rotation angle: +/- 75° from normal (top to bottom)
- Horizontal rotation angle: +/- 75° from normal (left to right)
- View time per unit:
 - Cosmetic front surface: 3seconds
 - Cosmetic edges: 3 seconds
 - Cosmetic back side: 3seconds

COSMETIC TEMPLATE-MEDIUM					
069-1770					
08/09/06 Rev A					
SIZE (mm ²)	DEFECT SHAPE	CLASS: A B C D			
		A	B	C	D
0.05		OK	OK	OK	Defects OK if no impact to mechanical performances
0.08					
0.1					
0.2					
0.3					
0.5					
0.7					
1.0					
1.5					
2.0					
NOTES	1) Max view time 5 seconds, viewing distance 50cm, illumination 2 parallel 32Watt 3500K fluorescent tubes at 1.5m distance. (View angle 45-90 degrees rotate part)				
	2) Defects Types: Scratch, Molded Internal Contamination, Black Dot/Line, Air Hole, Pit, Dent, Crack.				
	3) The spacing between any two flaws for front and top surfaces should be greater than 20mm.				
	4) Unless specified in cosmetic part drawing, use 25% contrast standard for accept/reject for visible hairline scratch (<0.001mm depth), knit line, gate blush and stress whitening.				
	5) Surface contamination that can NOT be cleaned off, Reject. Clean w/soft cotton cloth and NO isopropyl alcohol.				
	SCALE				
	CONTRAST STANDARD 25%				

• 613-35102/613-35106



Surface Mapping

Main Surface	Back Surface	Front Surface	Side Surface	Edge
M1/M2	B1	F1/F2	S1/S2	All edge

NO.	Defect Name	Location	Spec	Limit Sample Needed	APN Affected	Page #
1	Chip	All surface and Edge	1. Individual chip w/ plating: S>0.3mm^2: not allowed ; 0.2<S≤0.3mm^2, 1x allowed per edge ; 0.1<S≤0.2mm^2, 3x allowed per edge ; S≤0.1mm^2, no qty limitation . 2. Sawtooth chip(锯齿边)/Continuous chip w/ plating: S≤0.1mm^2, but total length<2mm, 1x allowed per edge ; S>0.1mm, not allowed . 3. Chip w/o plating, not allowed .	No	613-35102 613-35106	6
2	Bubble	All surface and Edge	Bubble S≤0.1mm^2, allowed . Bubble S>0.1mm^2, not allowed .	No		7
3	Scratch	All surface	1. Scratch with finger pulp feeling, not allowed . 2. Scratch without feeling: 4mm<L≤6mm, 1x allowed per surface ; 2.5≤L≤4mm, 3x allowed per surface ; L<2.5mm, no qty limitation .	No		8
4	Contamination	All surface	Removable, not allowed . Unremovable, if 25% grey film can cover, OK ; otherwise, NG .	No		9
5	Discoloration	All surface	In individual magnet, cover with 25% grey film to check, if covered, OK ; otherwise, NG . Between individual magnets, allowed .	No		10
6	Machining Mark	All surface	One line across whole surface, not allowed . Uniform lines across surface, allowed .	No		11
7	Pitting	All surface	S>0.1mm^2, not allowed . S≤0.1mm^2, allowed .	No		12
8	Magnet misalignment	M1 surface	If pass go gauge, OK ; otherwise, NG .	No		13

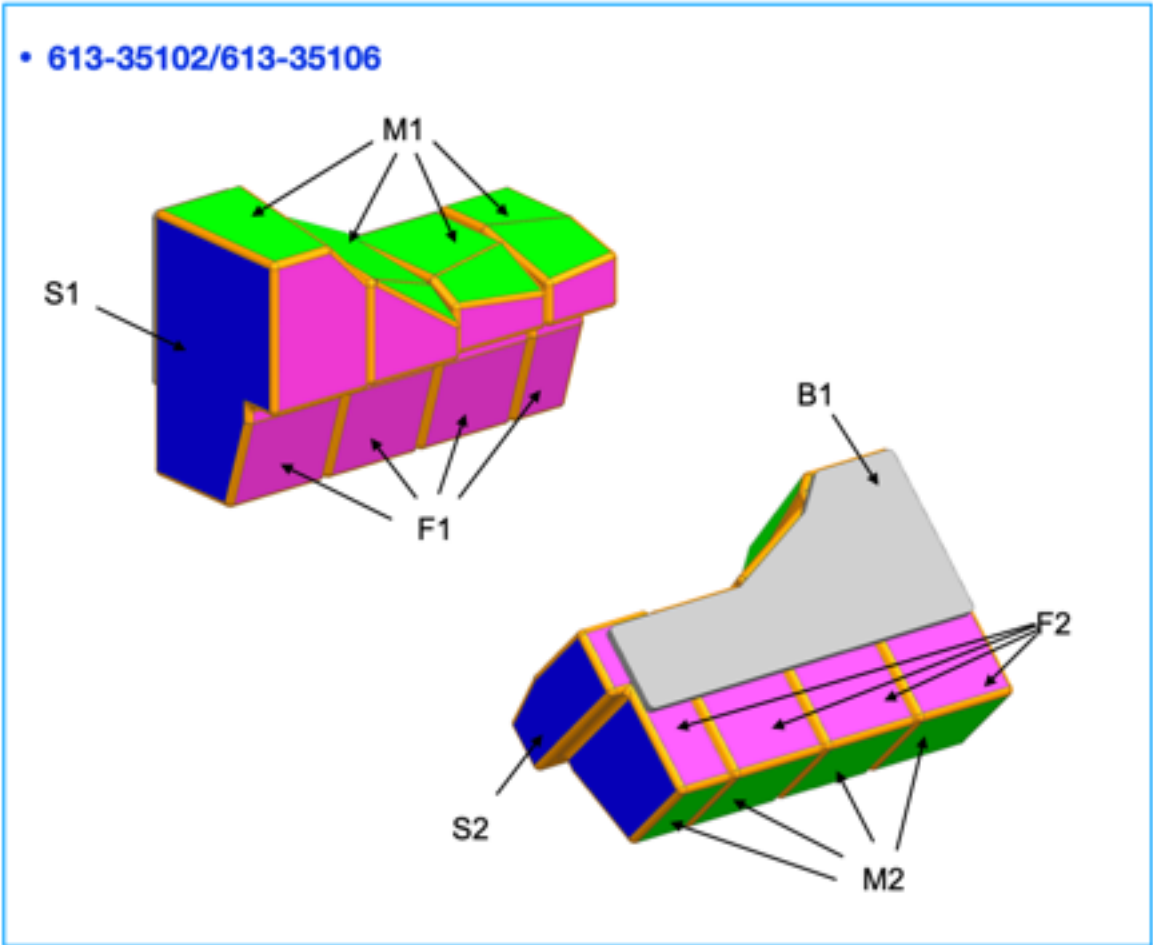
B747 Ramp LID Magnet & Shunt Cosmetic – CDML Index

Released on:2025-08-04

NO.	Defect Name	Location	Spec	Limit Sample Needed	APN Affected	Page #
9	Shunt Shifting	B1	All side, shunt not exceeded is allowed , otherwise, NG ;	No	613-35102 613-35106	14
10	Gap	M1/M2/B/F1/F2 surface	If pass 0.10mm plug gauge, NG ; otherwise, OK .	No		15
11	De-bonding	M1/M2/B/F1/F2 surface	Not allowed ;	No		16
12	Glue mark	M1/M2/F1/F2 surface	Glue mark after clean glue, Allowed ;	No		17
13	Glue Overflow	M1 surface B surface	M1 surface height maximum 0.1mm, allowed , otherwise, not allowed . B surface, glue height not exceed Shunt, allowed , otherwise, not allowed .	No		18
14	Shunt Deform	B surface	If can pass surface nogo gauge, allowed, otherwise, not allowed .	No		19
15	Wrong Polarity	F surface	Wrong polarity, not allowed .	No		20
16	Weak magnetism	F surface	Not allowed .	No		21
17	Wrong orientation	F surface	Wrong orientation, not allowed .	No		22

B747 Ramp BUD Magnet Cosmetic – Chip

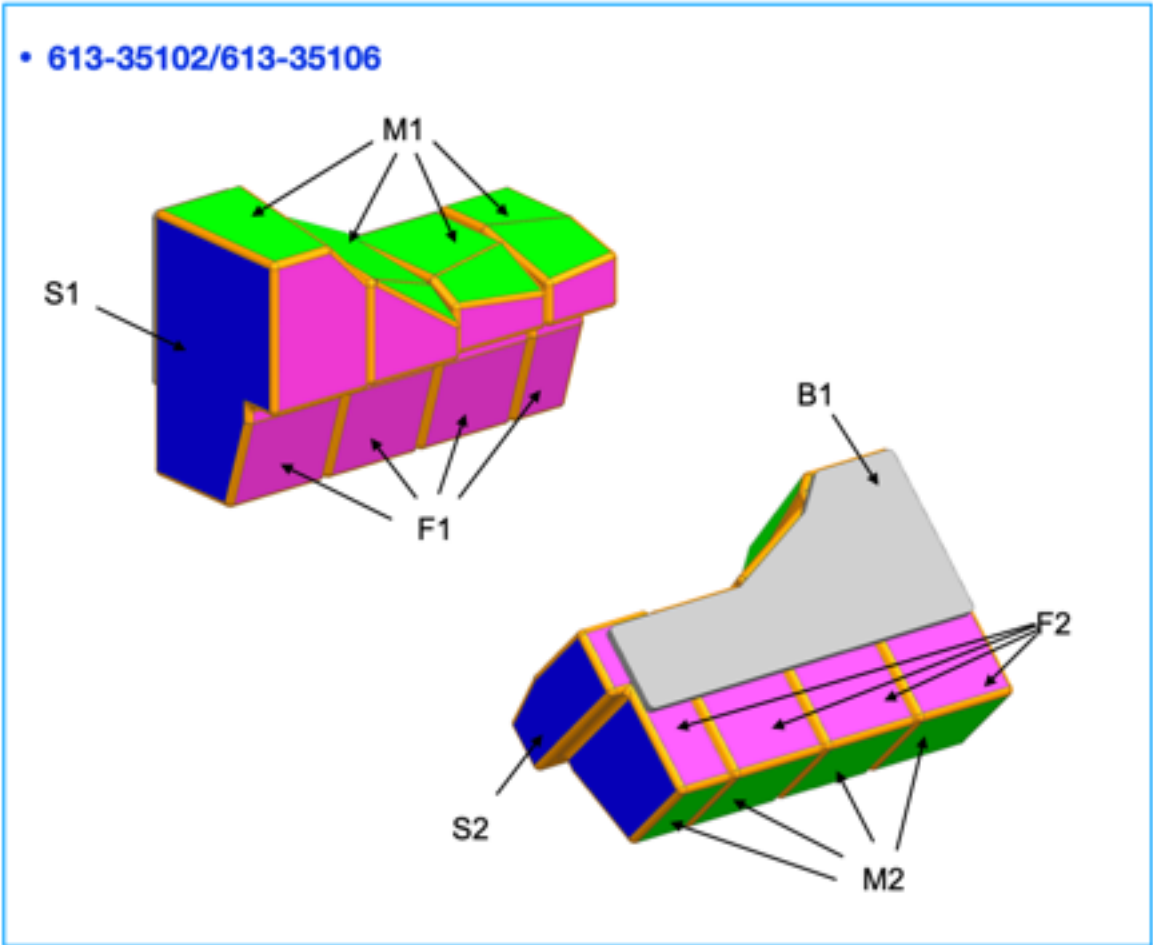
- Defect Name: Chip
- Location: All surface and Edge
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
1. Individual chip w/ plating: $S > 0.3\text{mm}^2$: not allowed; $0.2 < S \leq 0.3\text{mm}^2$, 1x allowed per edge; $0.1 < S \leq 0.2\text{mm}^2$, 3x allowed per edge; $S \leq 0.1\text{mm}^2$, no qty limitation. 2. Sawtooth chip/Continuous chip w/ plating: $S \leq 0.1\text{mm}^2$, but total length $< 2\text{mm}$, 1x allowed per edge; $S > 0.1\text{mm}$, not allowed. 3. Chip w/o plating, not allowed.	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 $< 0.1\text{mm}^2$, no qty limitation  $0.1 < S \leq 0.2\text{mm}^2$, 3x per surface allowed  $0.2 < S \leq 0.3\text{mm}^2$, 1x per surface allowed	/	 Chip $S > 0.3\text{mm}^2$, not allowed				

B747 Ramp BUD Magnet Cosmetic – Bubble

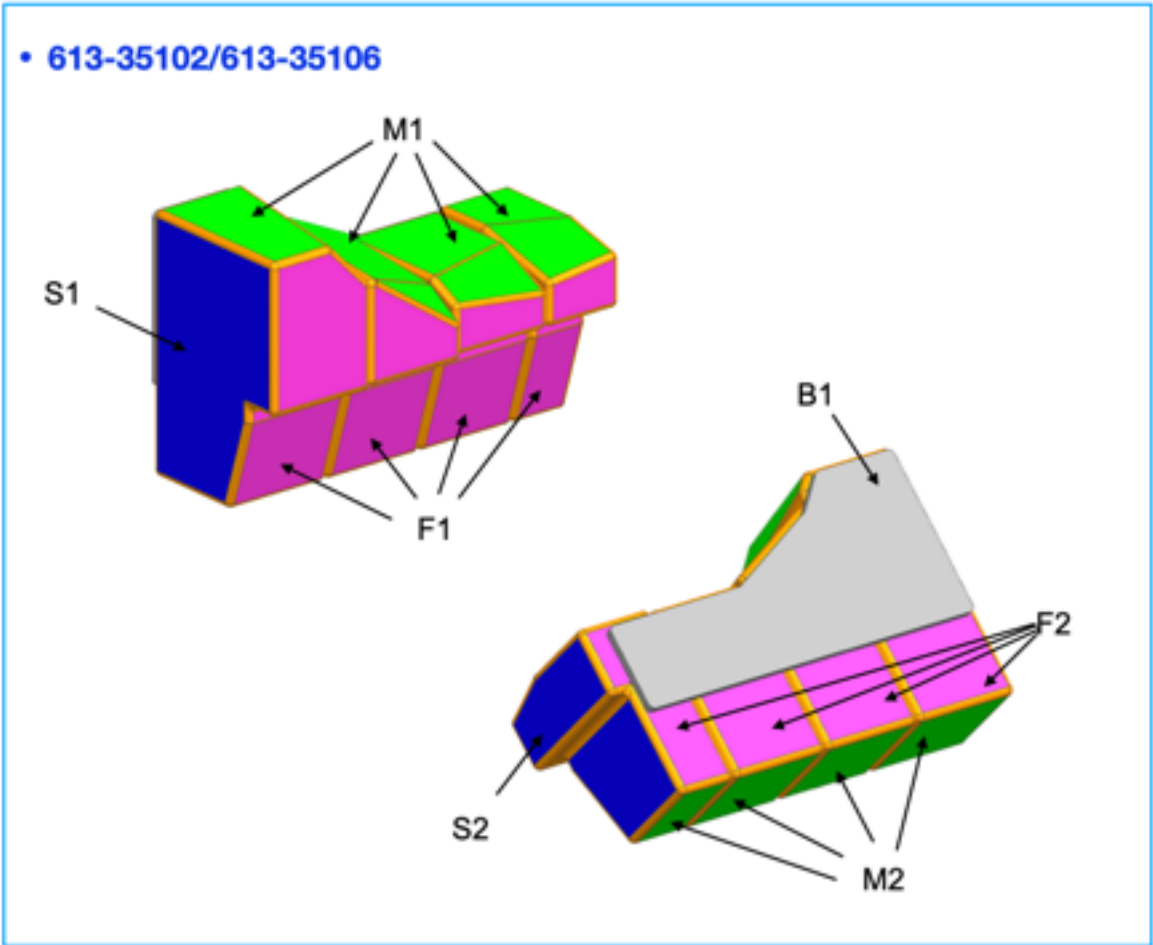
- Defect Name: Bubble
- Location: All surface and Edge
- APN Affected: 613-35102
613-35106

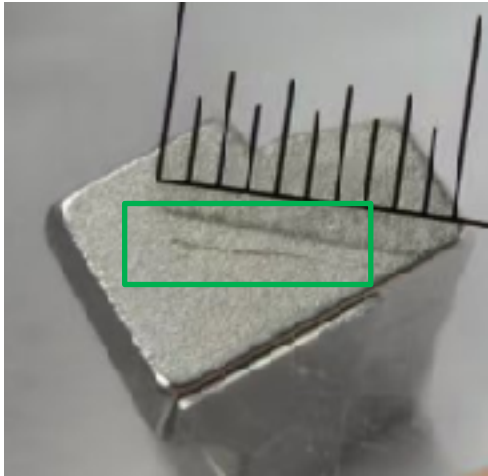
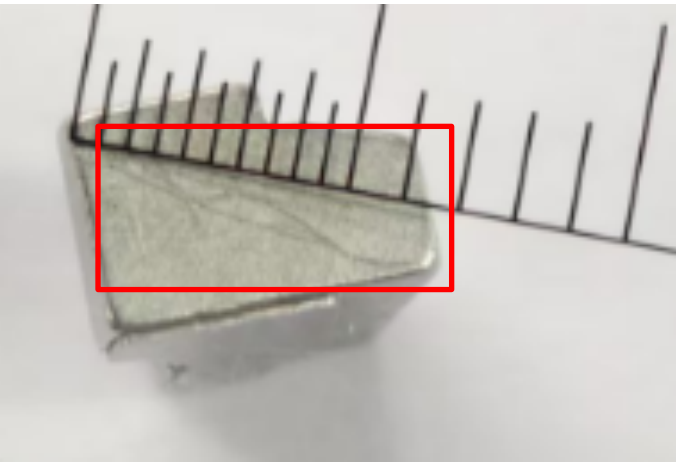


Spec	Alert Level	Surface Mapping				
Bubble $S \leq 0.1\text{mm}^2$, allowed. Bubble $S > 0.1\text{mm}^2$, not allowed.	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
	/					
Bubble $S \leq 0.1\text{mm}^2$, allowed						
		Bubble $S > 0.1\text{mm}^2$, not allowed				

B747 Ramp BUD Magnet Cosmetic – Scratch

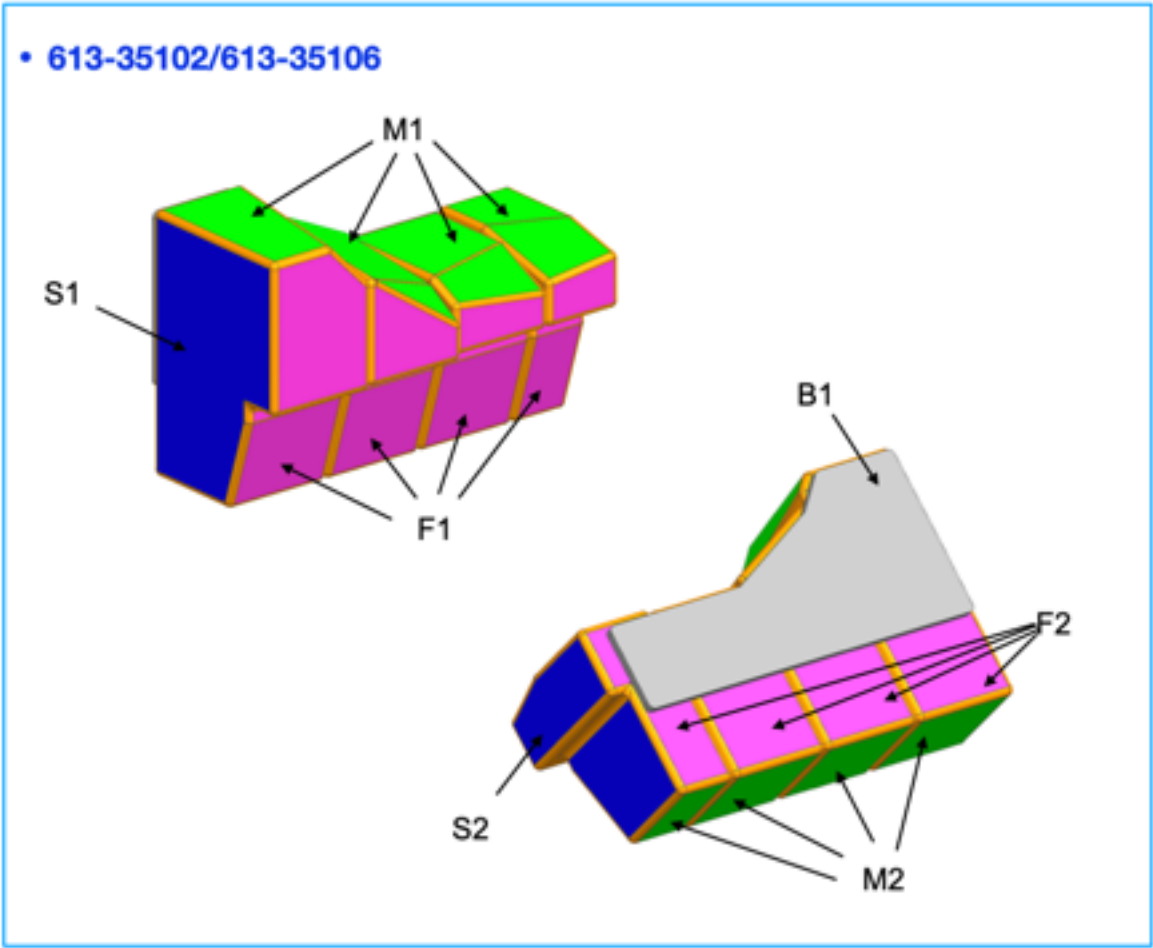
- Defect Name: Scratch
- Location: All surface
- APN Affected: 613-35102
613-35106

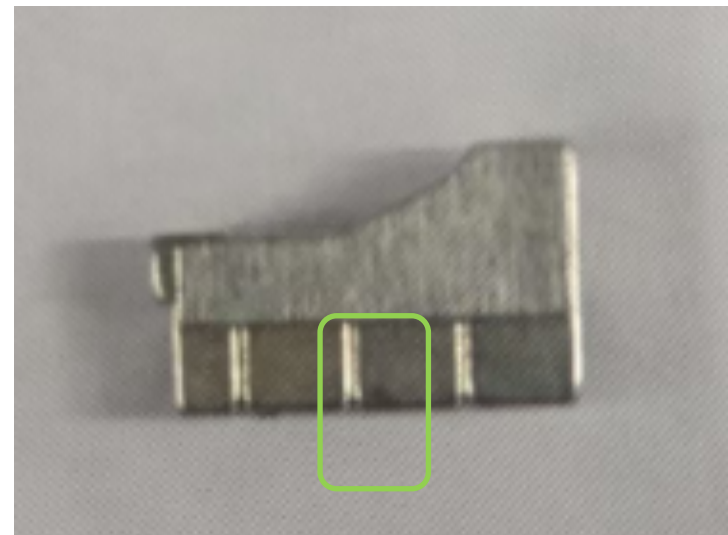
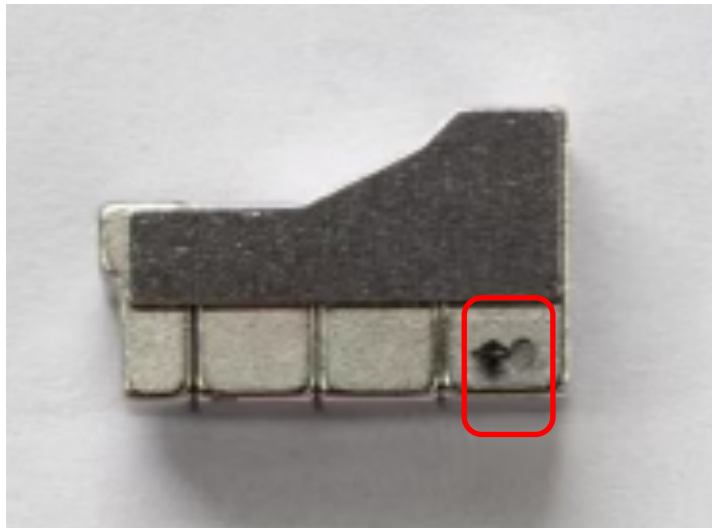
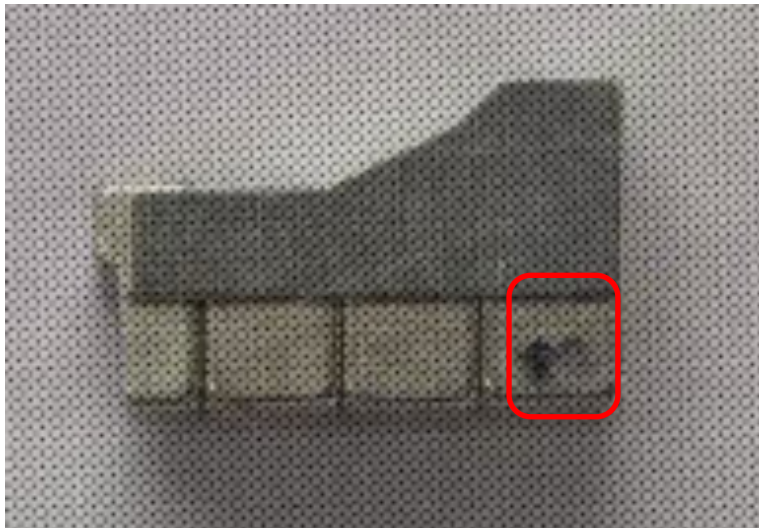


Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
1. Scratch with finger pulp feeling, not allowed . 2. Scratch without feeling: 4mm<L≤6mm, 1x allowed per surface; 2.5≤L≤4mm, 3x allowed per surface; L<2.5mm, no qty limitation.	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 	/	 Scratch L>6mm, not allowed				

B747 Ramp BUD Magnet Cosmetic – Contamination

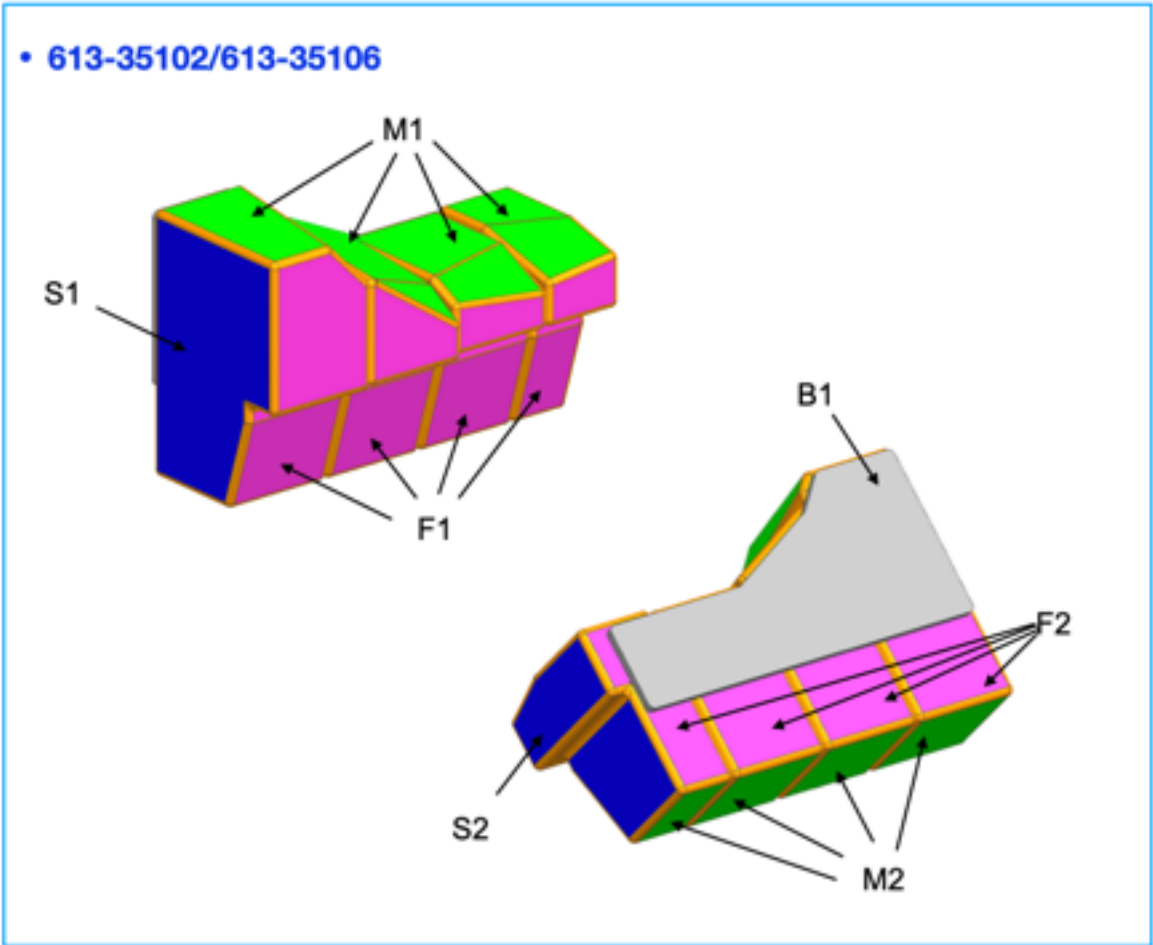
- Defect Name: Scratch
- Location: All surface
- APN Affected: 613-35102
613-35106

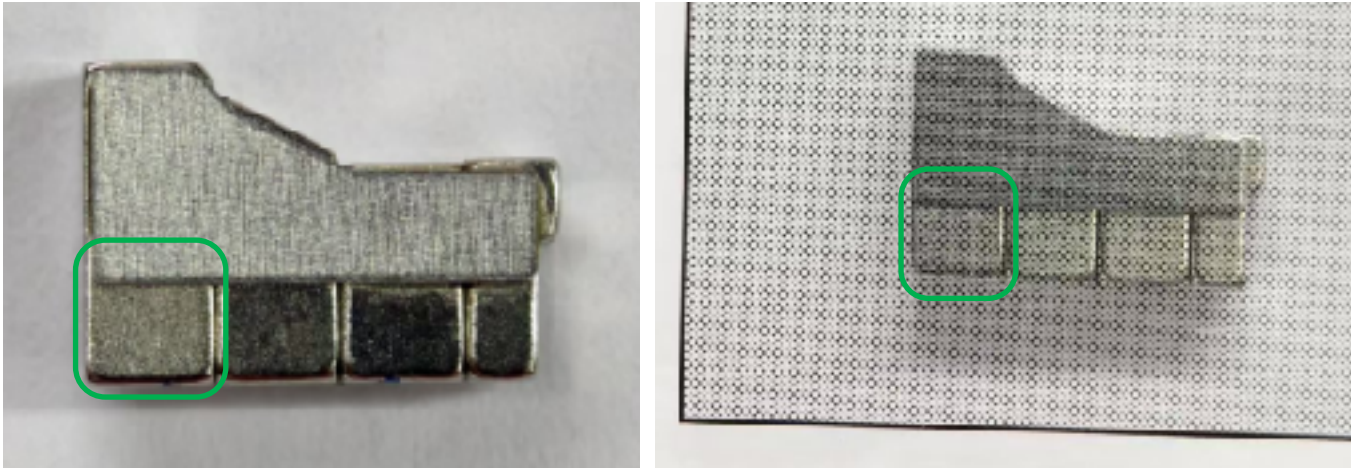


Spec	Alert Level	Surface Mapping				
Removable, not allowed . Unremovable, if 25% grey film can cover, OK ; otherwise, NG .	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
  Unremovable, cover with 25% grey film, allowed	/	  Unremovable, can't be covered with 25% grey film, not allowed				

B747 Ramp BUD Magnet Cosmetic – Discoloration

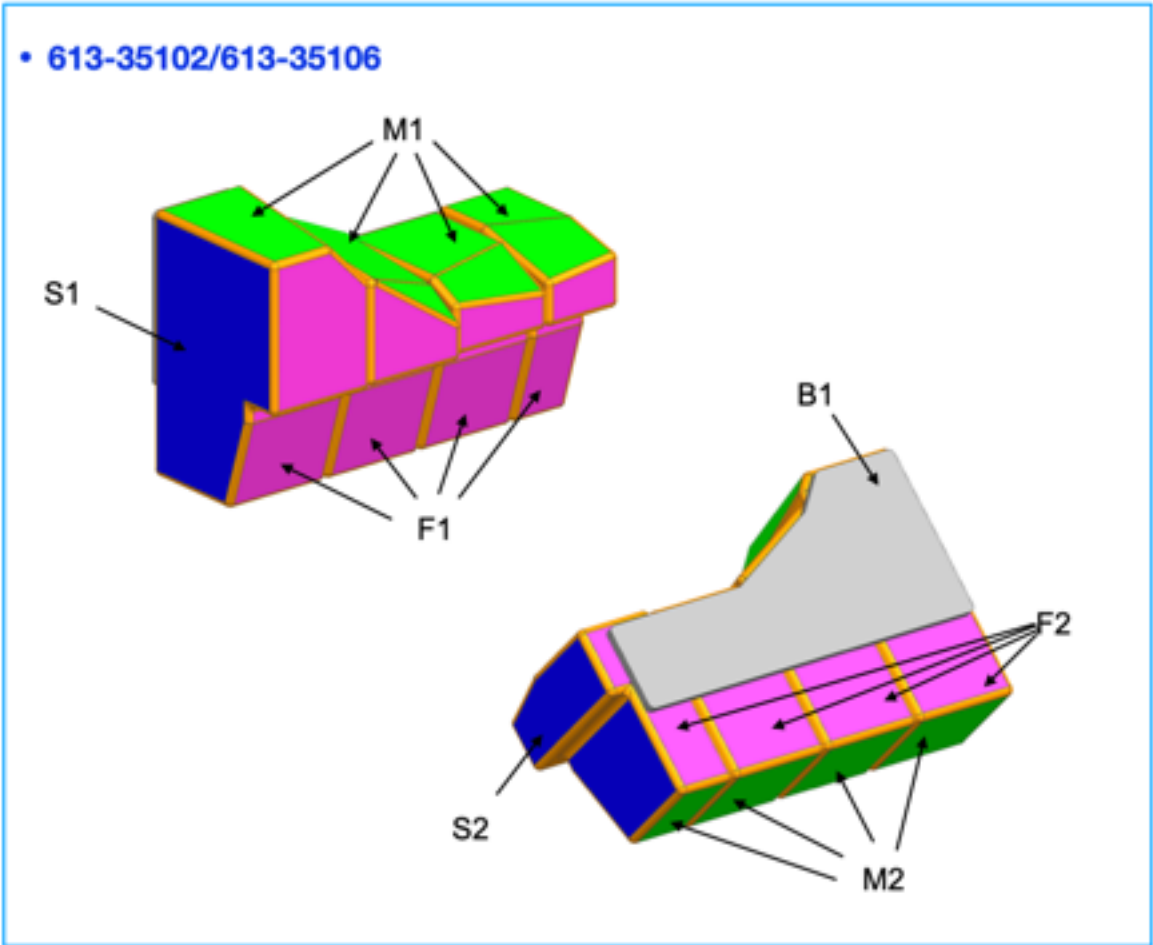
- Defect Name: Discoloration
- Location: All surface
- APN Affected: 613-35102
613-35106

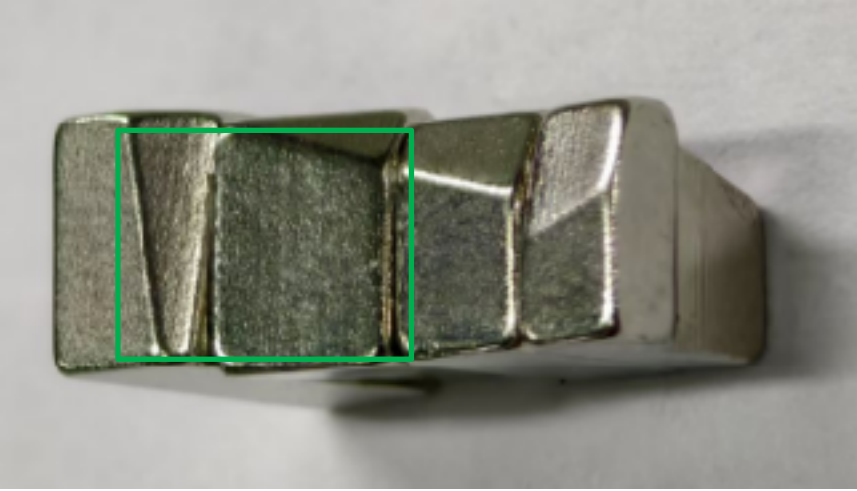


Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
1. In individual magnet, cover with 25% grey film to check, if covered, OK; otherwise, NG. 2. Between individual magnets, allowed.	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 In individual magnet, cover with 25% grey film, allowed	/	/				

B747 Ramp BUD Magnet Cosmetic – Machining Mark

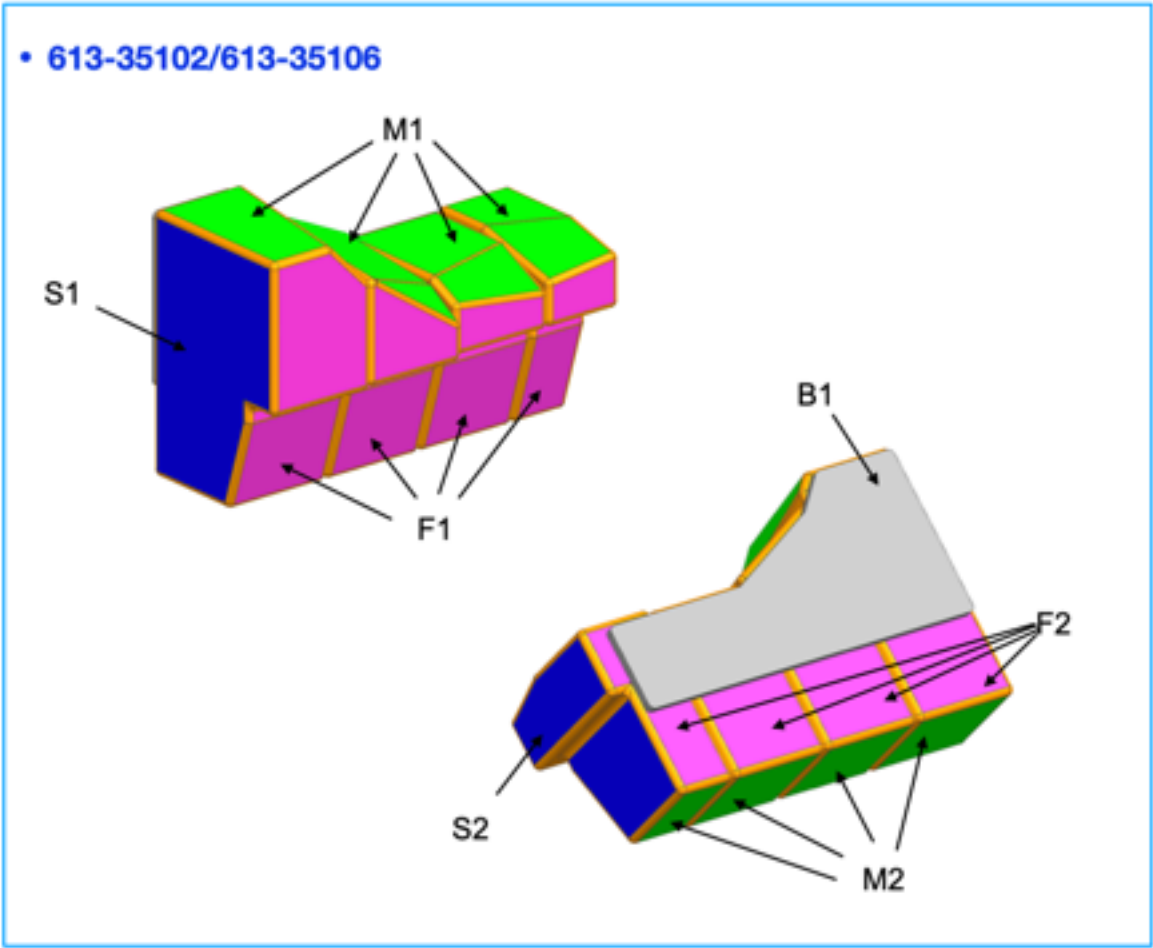
- Defect Name: Machining Mark
- Location: All surface
- APN Affected: 613-35102
613-35106

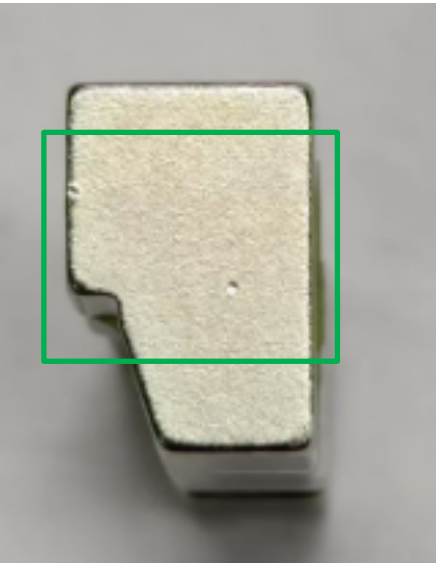
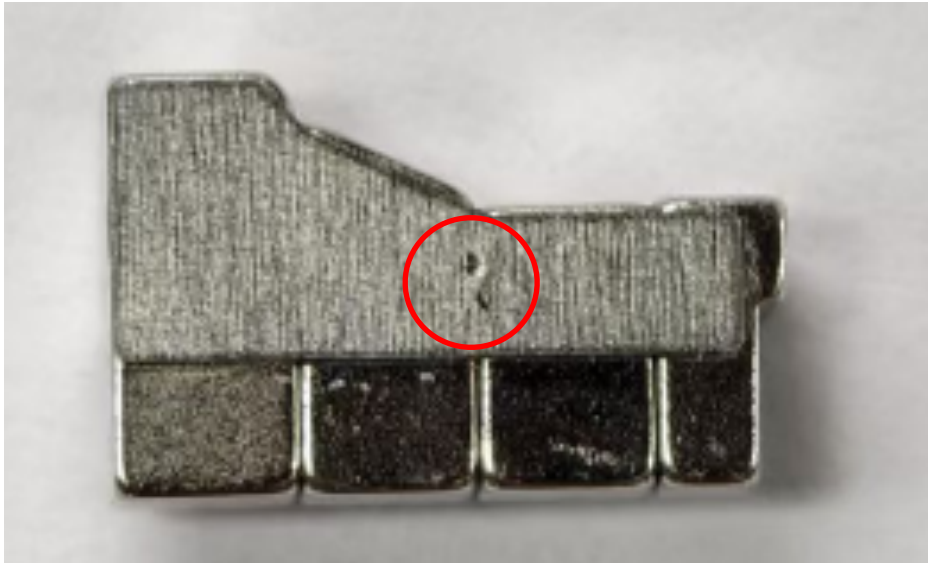


Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
One line across whole surface, not allowed . Uniform lines across surface, allowed .	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 W/O feeling: Multiple lines across surface, allowed	/	 W/O feeling: One line across whole surface, not allowed				

B747 Ramp BUD Magnet Cosmetic – Pitting

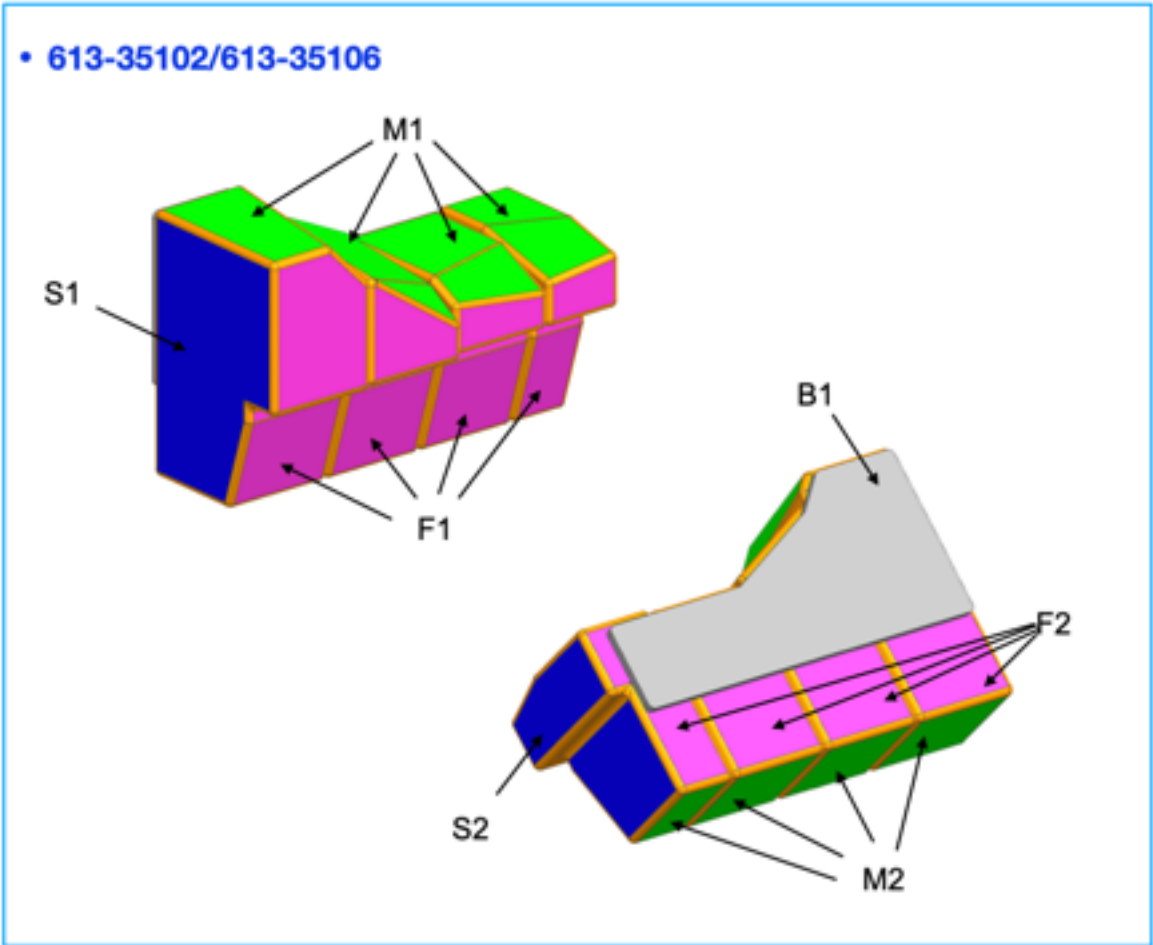
- Defect Name: Pitting
- Location: All surface
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
S>0.1mm^2, not allowed. S≤0.1mm^2, allowed.	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 Pitting S≤0.1mm^2, allowed	/	 Pitting S>0.1mm^2, not allowed				

B747 Ramp BUD Magnet Cosmetic – Magnet misalignment

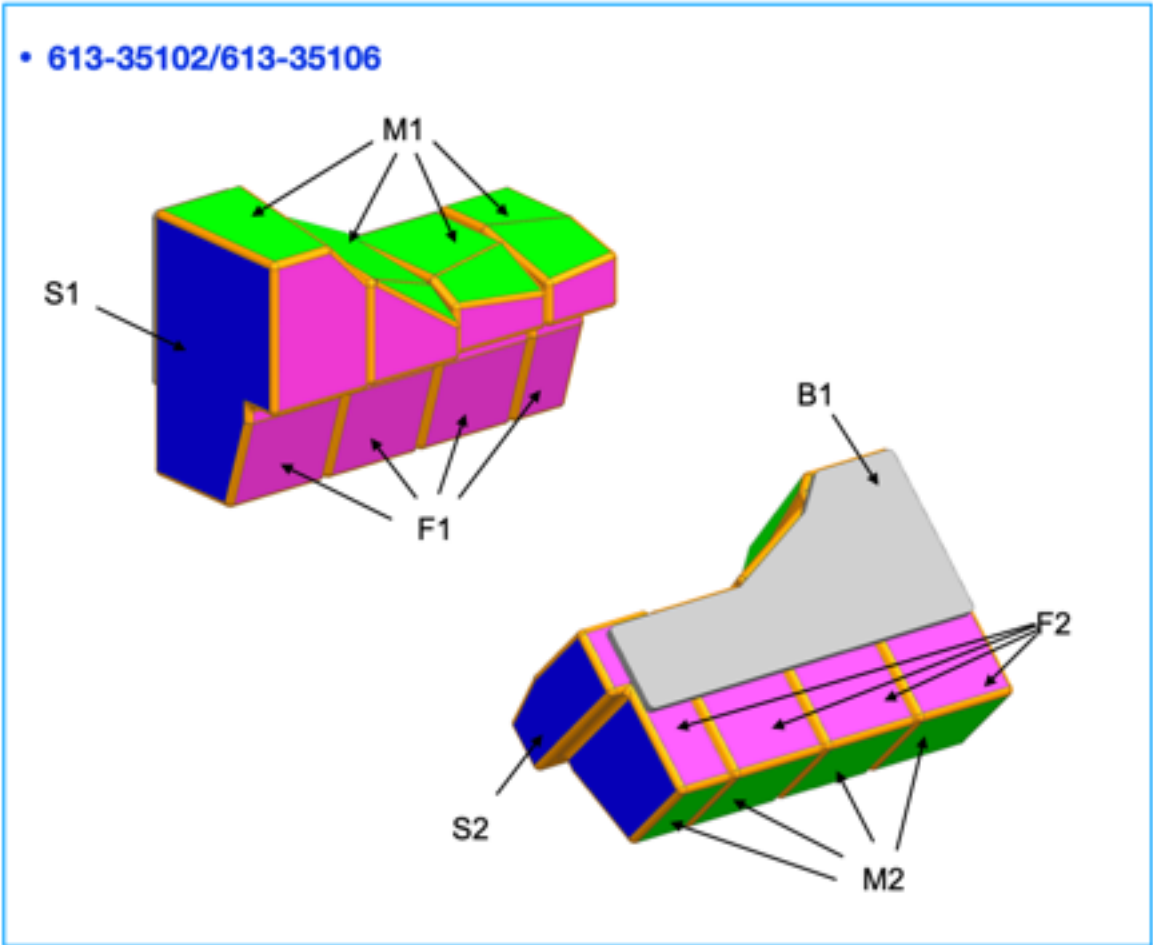
- Defect Name: Magnet misalignment
- Location: All surface
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
If pass go gauge, OK; otherwise, NG.	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
If pass go gauge, allowed Go gauge: ①②③ Nogo gauge: ④(reference) Length go gauge Height go gauge Width go gauge	/	If Go-no go gauge can not pass, not allowed				

B747 Ramp BUD Magnet Cosmetic – Shunt Shifting

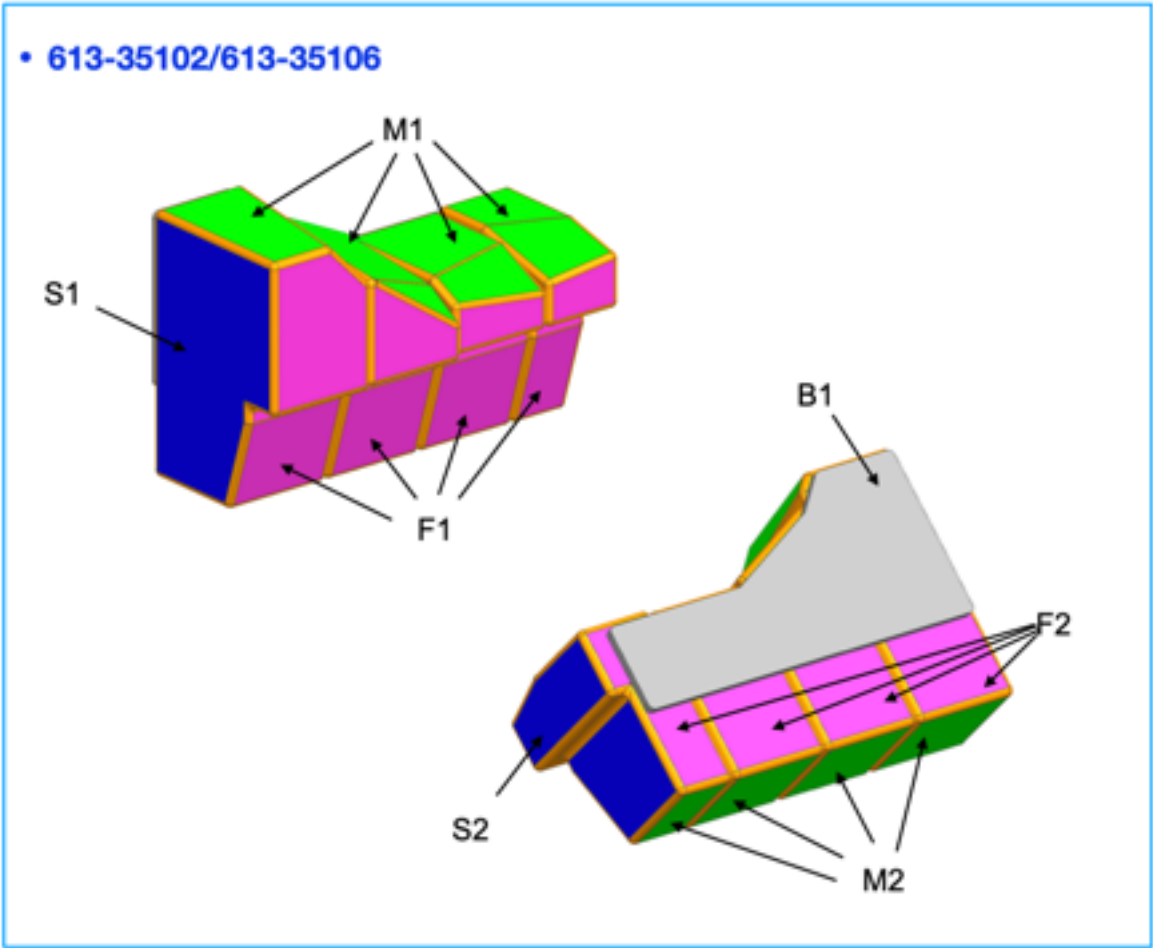
- Defect Name: Shunt Shifting
- Location: B1
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
All side, shunt not exceeded is allowed, otherwise, NG;	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
	/					

B747 Ramp BUD Magnet Cosmetic – Gap

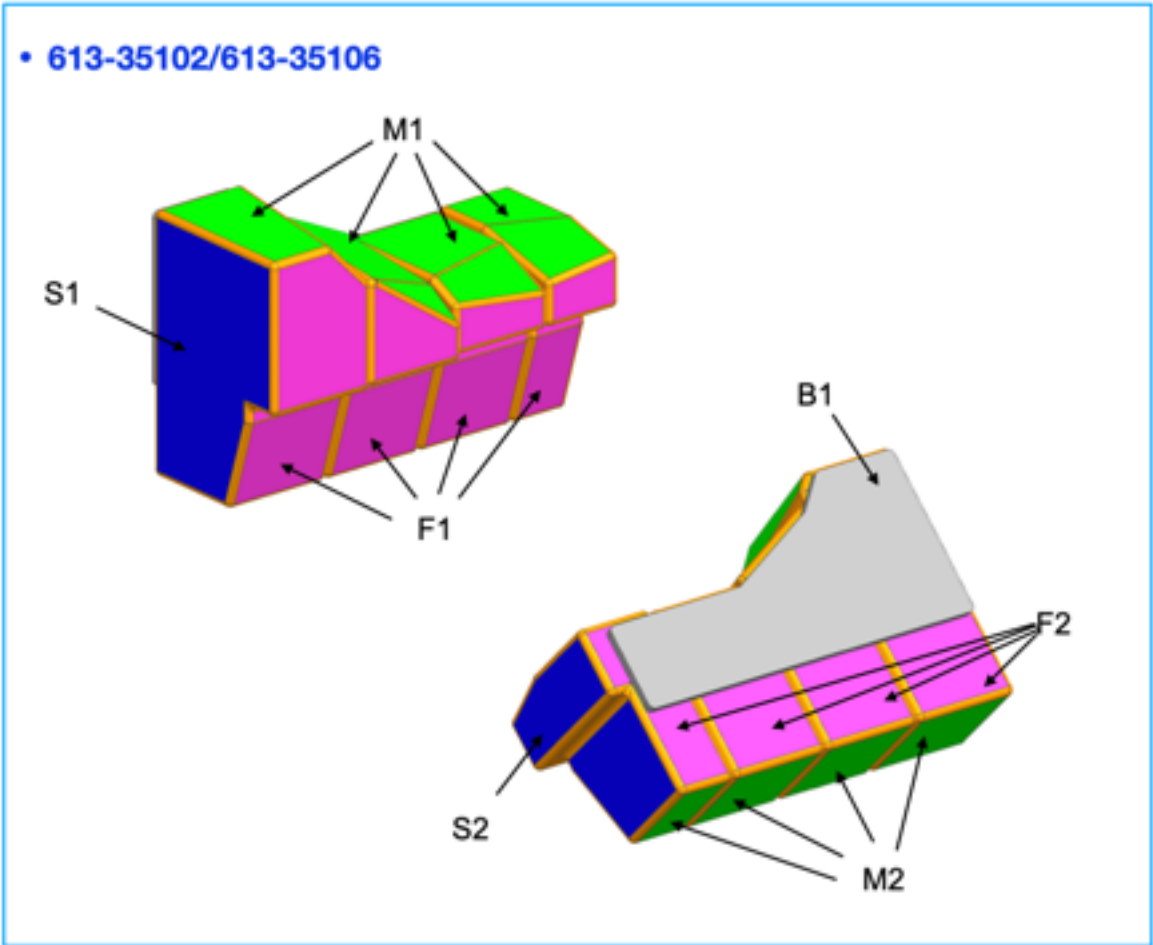
- Defect Name: Gap
- Location: M1/M2/B/F1/F2 surface
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
If pass 0.10mm plug gauge, NG; otherwise, OK.	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
Normal sample	/	Pass 0.10mm plug gauge, not allowed				

B747 Ramp BUD Magnet Cosmetic – De-bonding

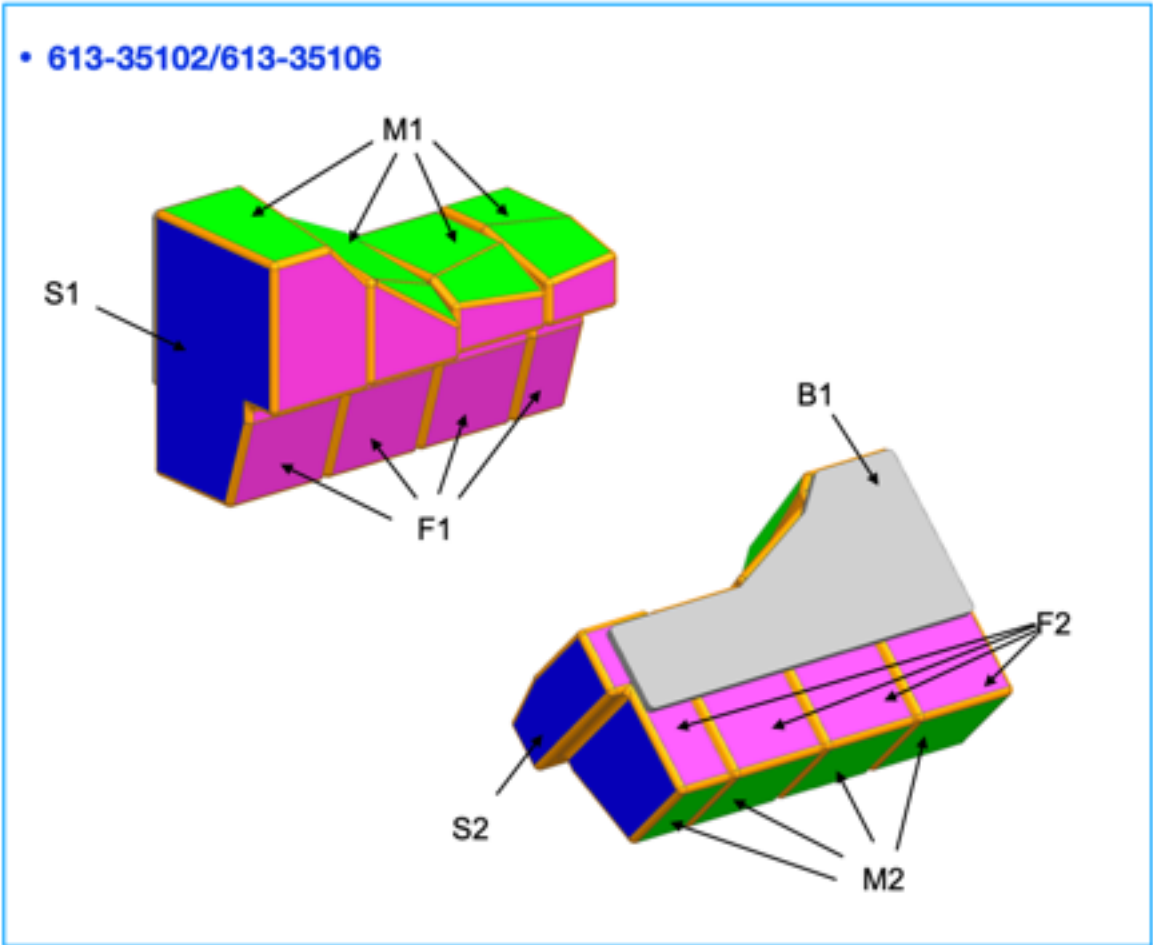
- Defect Name: De-bonding
- Location: M1/M2/B/F1/F2 surface
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
Not allowed;	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
	/					
Normal sample		Magnet or shunt de-bonding, not allowed				

B747 Ramp BUD Magnet Cosmetic – Glue mark

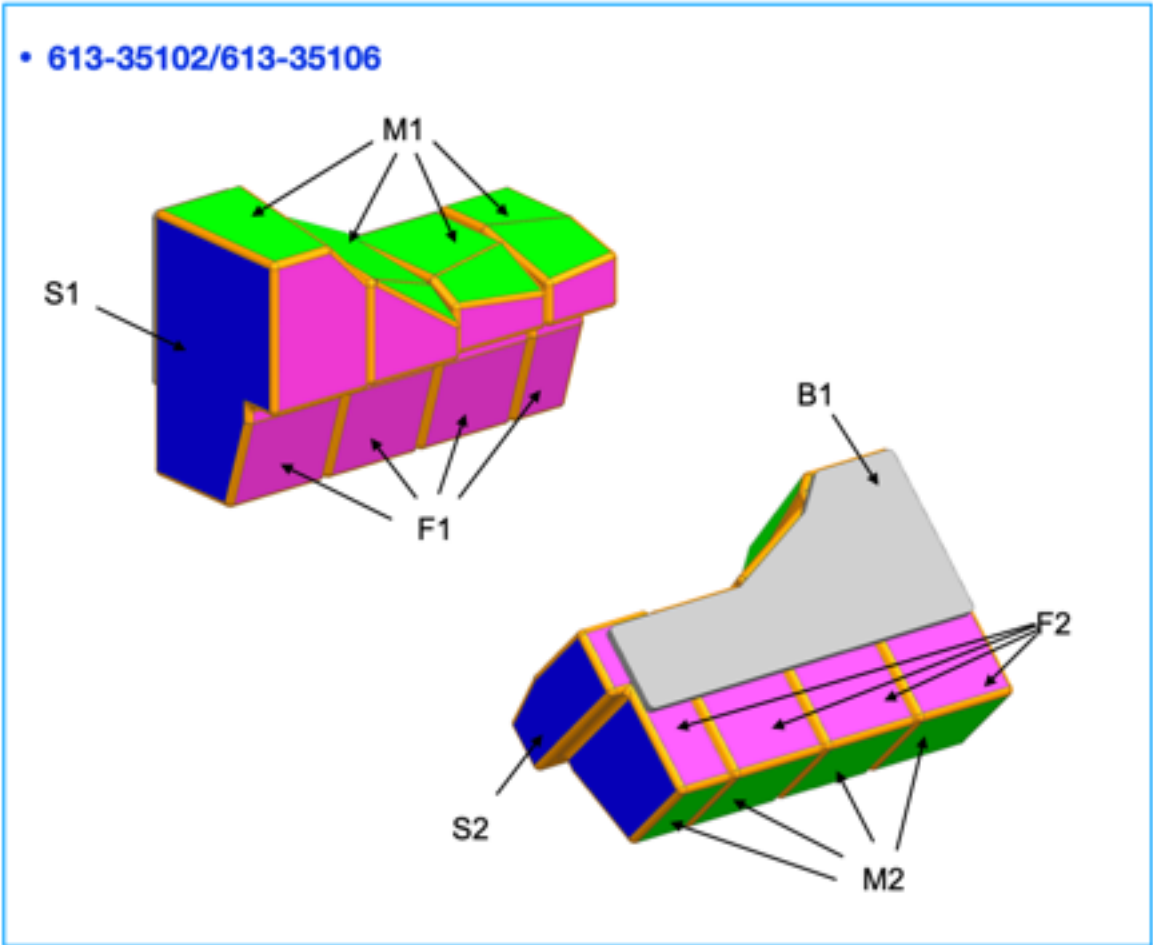
- Defect Name: Glue mark
- Location: M1/M2/B/F1/F2 surface
- APN Affected: 613-35102
613-35106

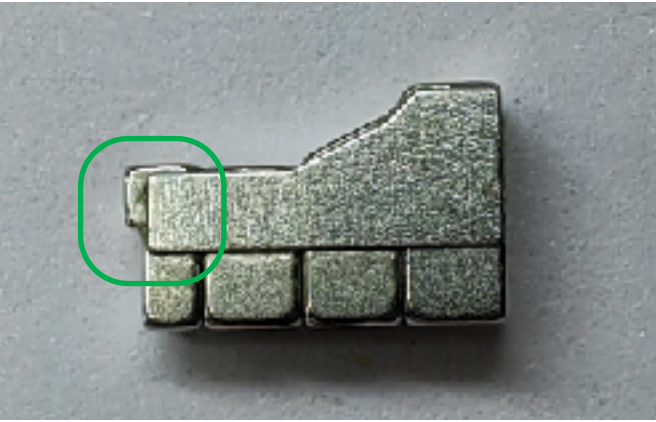
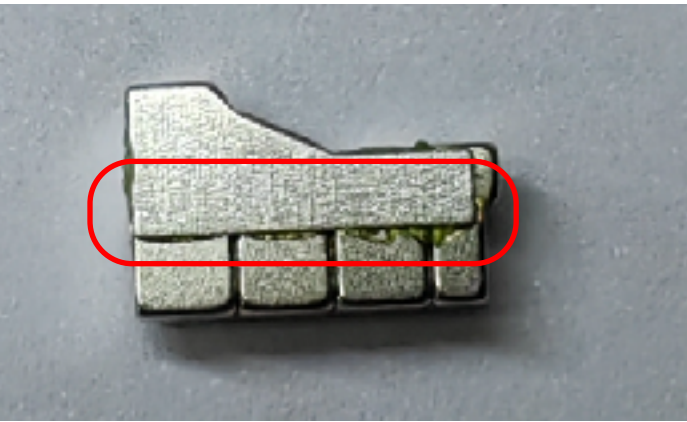


Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
Glue mark after clean glue, Allowed;	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 Glue mark after glue cleaning, no dimension impact, allowed	/	 w/o glue cleaning, not allowed				

B747 Ramp BUD Magnet Cosmetic – Glue Overflow

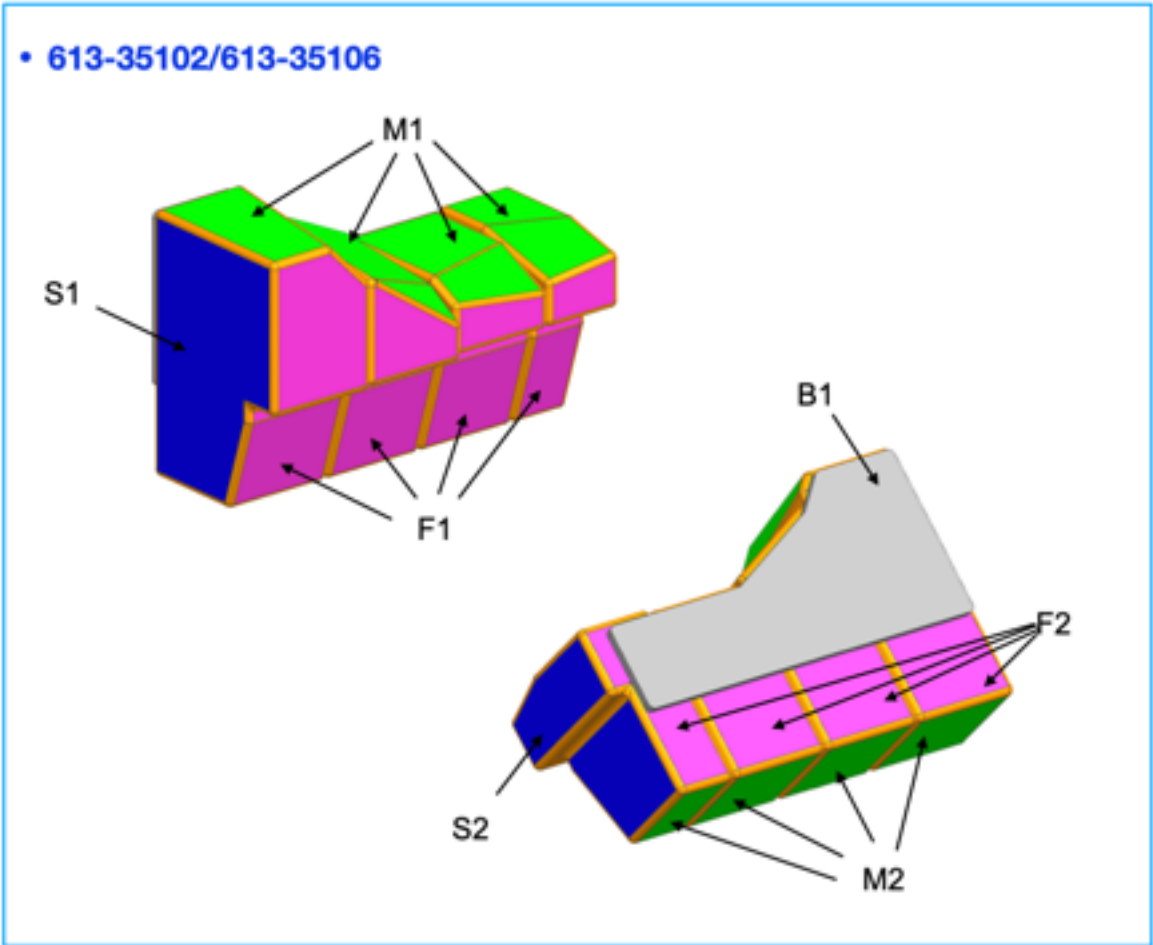
- Defect Name: Glue Overflow
- Location: M1 surface/B surface
- APN Affected: 613-35102
613-35106

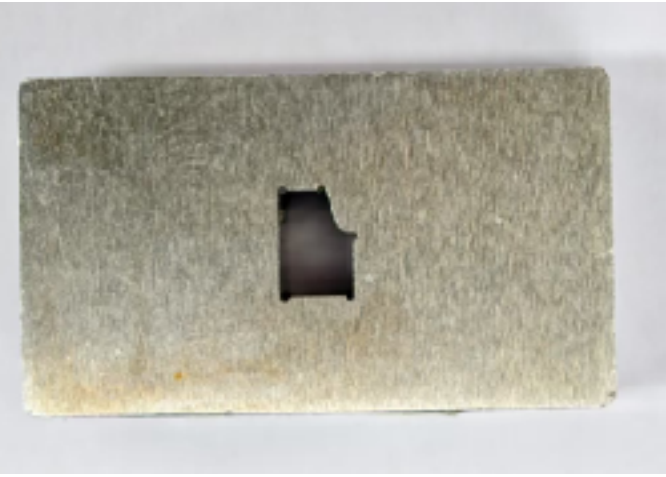


Spec	Alert Level	Surface Mapping				
M1 surface height maximum 0.1mm, allowed , otherwise, not allowed . B surface , glue height not exceed Shunt, allowed , otherwise, not allowed .	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 Glue height not exceed Shunt, allowed	/	 Glue height exceed Shunt, not allowed				

B747 Ramp BUD Magnet Cosmetic – Shunt Deform

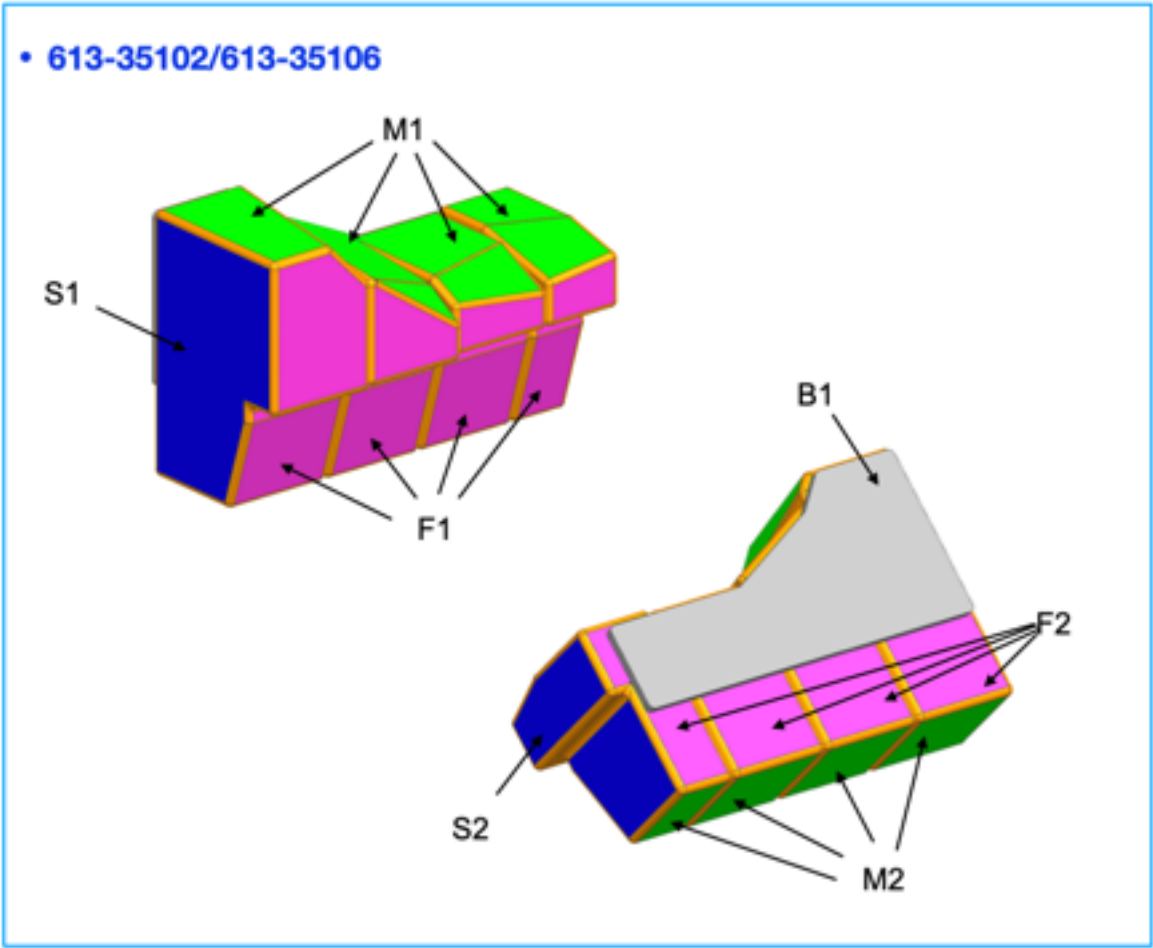
- Defect Name: Shunt Deform
- Location: B surface
- APN Affected: 613-35102
613-35106

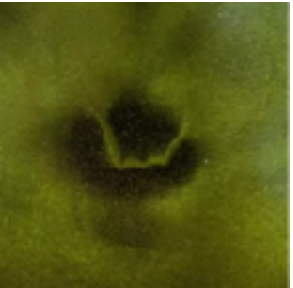
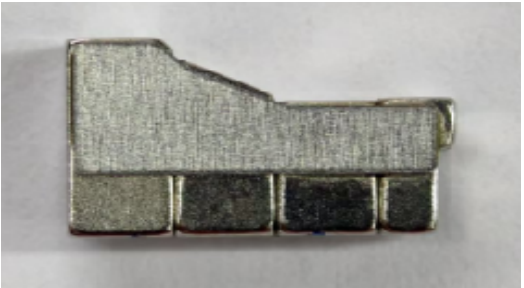
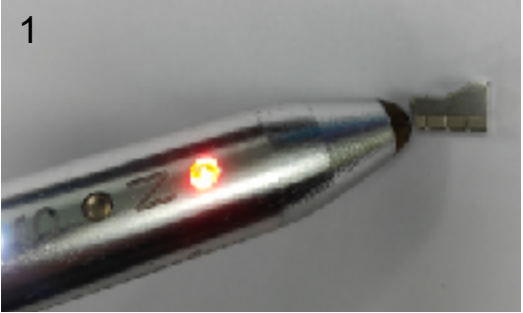
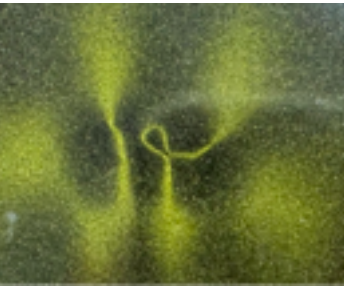


Spec	Alert Level	Surface Mapping				
If can pass surface nogo gauge, allowed, otherwise, not allowed.	Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
		M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
	/	 If can pass surface no go gauge, allowed, otherwise, not allowed;				

B747 Ramp BUD Magnet Cosmetic – Wrong Polarity

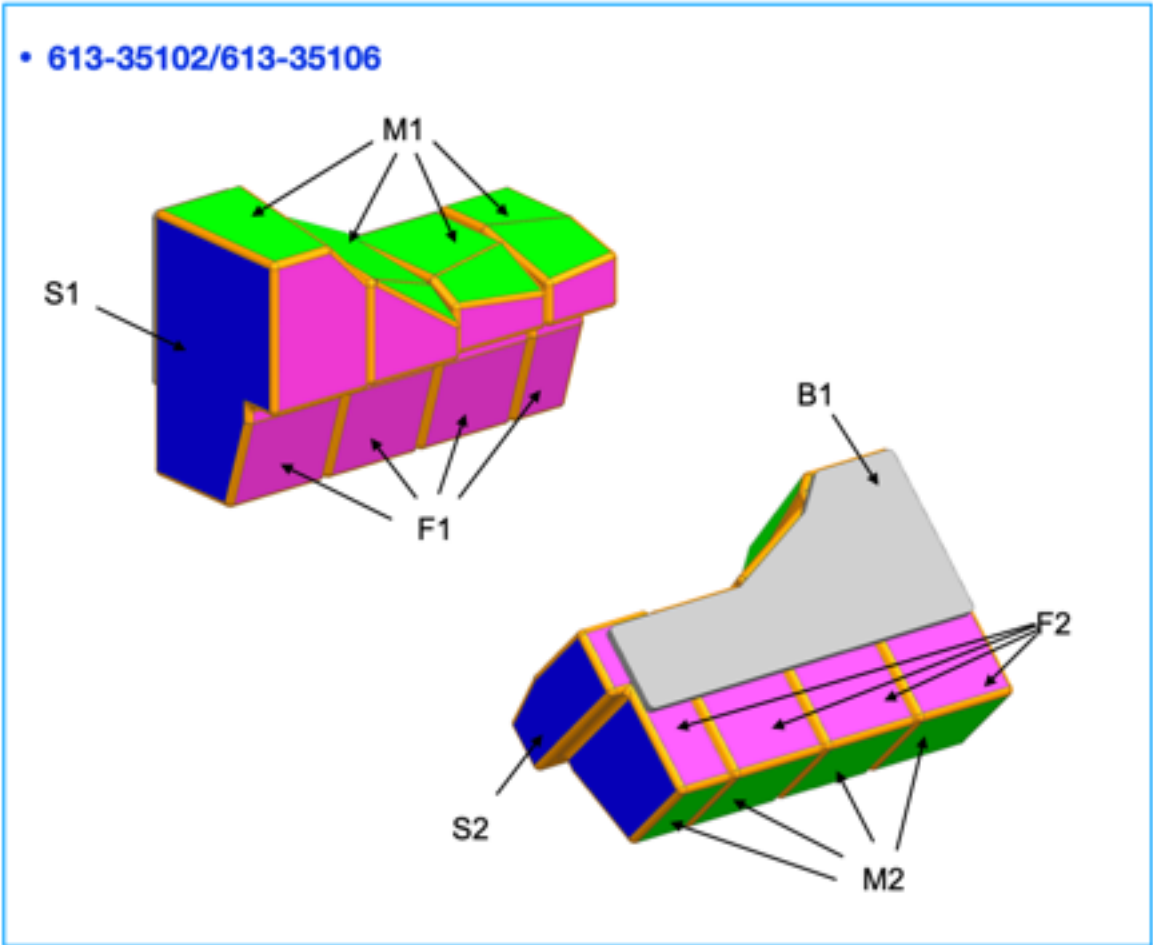
- Defect Name: Wrong Polarity
- Location: F surface
- APN Affected: 613-35102
613-35106

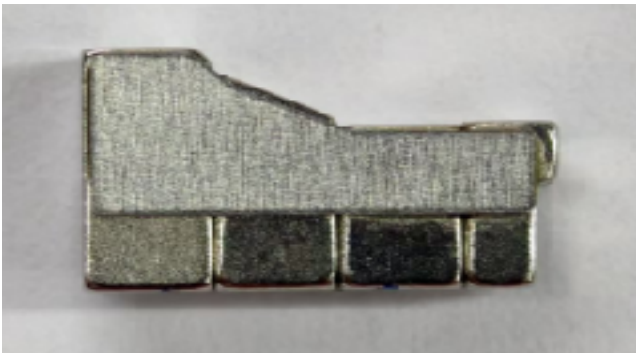
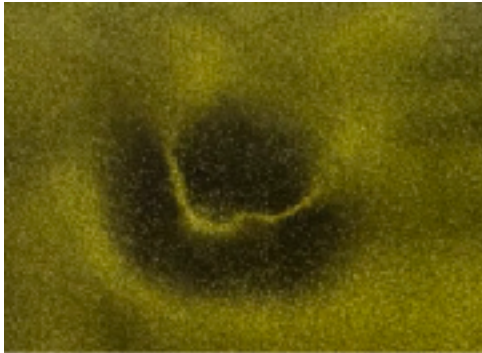


Spec		Alert Level	Surface Mapping				
Wrong polarity, not allowed.		Moderate	Main Surface	Back Surface	Front Surface	Side Surface	Edge
			M1/M2	B1	F1/F2	S1/S2	All edge
OK		/	NG				
 1st step, cover with magnetic film 2nd step, verify the profile(like gold ingot)	 1 3  2 4	/	 Wrong polarity, not allowed.				
3rd, check the polarity 1-2-3-4							

B747 Ramp BUD Magnet Cosmetic – Weak magnetism

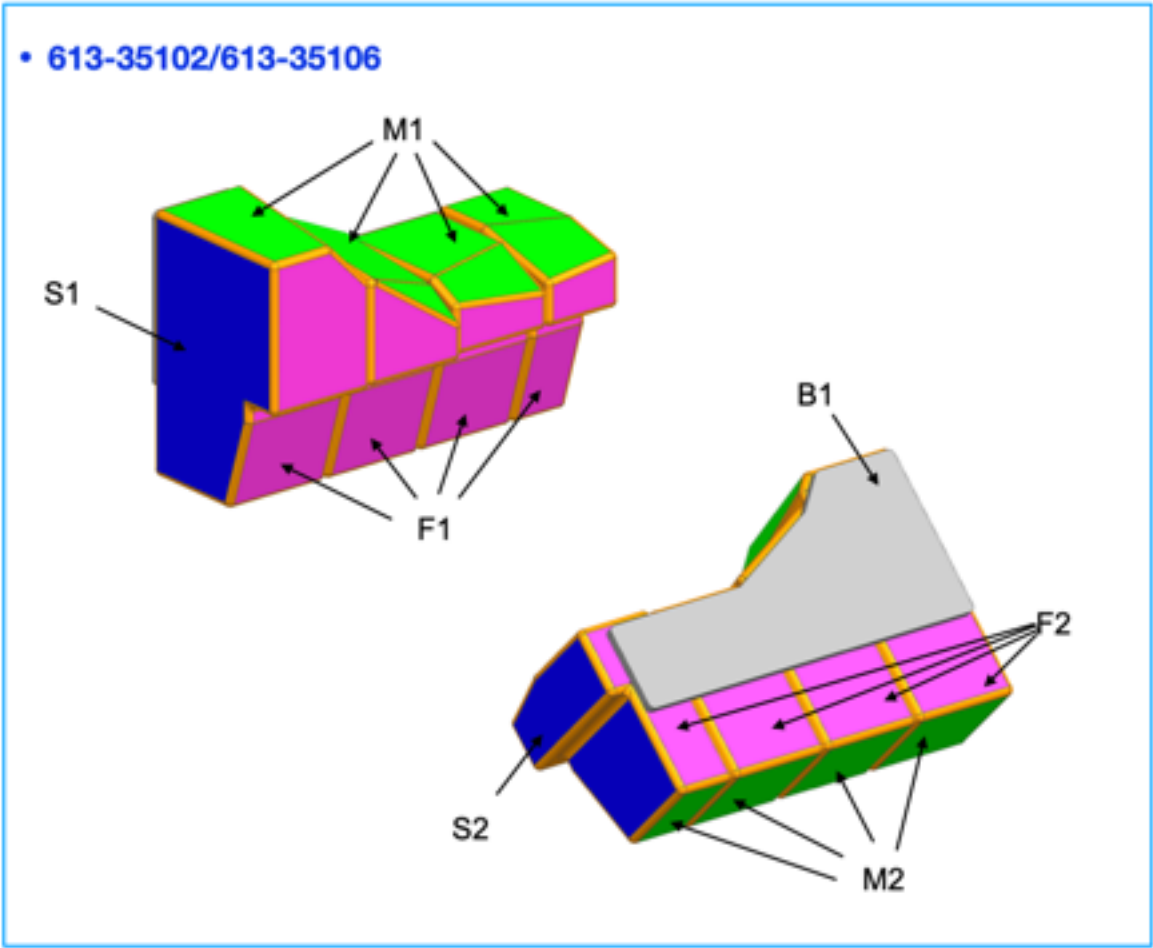
- Defect Name: Weak magnetism
- Location: F surface
- APN Affected: 613-35102
613-35106

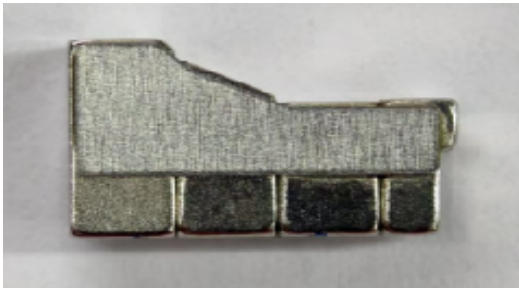
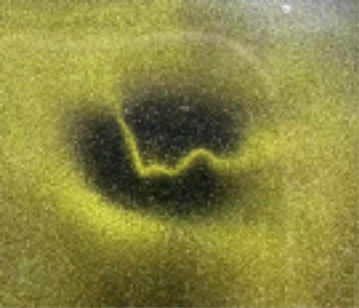


Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
Not allowed.	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 OK Sample	/	 Weak magnetism, not allowed.				

B747 Ramp BUD Magnet Cosmetic – Wrong orientation

- Defect Name: Wrong Polarity
- Location: F surface
- APN Affected: 613-35102
613-35106



Spec	Alert Level	Surface Mapping				
		Main Surface	Back Surface	Front Surface	Side Surface	Edge
Wrong orientation, not allowed .	Moderate	M1/M2	B1	F1/F2	S1/S2	All edge
OK	/	NG				
 OK Sample	/	 Wrong orientation, not allowed.				